

A Thermal-First Detection Cascade for Real-Time Human Search and Rescue on a 1 GB Embedded UAV Payload

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Payload firmware, datasets and evaluation scripts: github.com/saresq · commit a6b8d47

Abstract—Aerial search and rescue needs a sensor that works at night and a processor that fits on a small battery-powered airframe. These two requirements are usually in tension: the radiometric thermal cores used in the literature cost more than the rest of the aircraft, and the multispectral fusion networks that exploit them need a desktop GPU. We present SaResQ, a search-and-rescue payload built entirely from commodity parts around a Raspberry Pi 4 (1 GB) and a 32×24 MLX90640 thermal array, and show that a *nominate-then-confirm* cascade closes the gap. A non-learned robust z-score test over the thermal frame—median and median absolute deviation rather than mean and standard deviation, so that a warm body cannot inflate the threshold meant to find it—nominates two to six candidate regions per frame at negligible cost. Only those regions are passed to a YOLOv8-nano detector, which therefore performs between 1/16 and 1/3 of the arithmetic of a whole-frame detector. Candidates are projected into the visible camera through a six-parameter affine registration fitted by RANSAC over 38 automatically-detected correspondences, achieving a reprojection residual of **0.506 thermal pixels**, 7.4× better than the nominal scaling it replaces. A MobileNetV2 scene classifier supplies disaster context at 95.2% accuracy. Candidates are carried across frames by an ego-motion-compensated SORT tracker—a constant-velocity Kalman filter associated by the Hungarian algorithm, whose predictions are first displaced by a similarity transform derived from GNSS and gyroscope, without which a 95 px-per-frame drift at survey speed fragments every track—projected to latitude and longitude through a complementary-filter attitude estimate, and maintained on the ground as alpha-beta map tracks with Kalata gains and a χ^2 validation gate. A log-odds ledger accumulates evidence across repeated passes over the same ground position. On the target hardware the detector runs in **28.2 ms** per 160 px crop and the complete six-candidate worst case consumes 177.5 ms of a 250 ms frame budget, a 29% margin, in 95.2 MB of resident memory. The shipped detector attains Person AP₅₀ of **0.851** on HIT-UAV and **0.478** on RGBTDronePerson-thermal. We contrast this against a 2018 result on comparable hardware that reported 0.18 FPS for embedded CNN pedestrian detection, and against current RGBT fusion networks that require 22× the parameters and a discrete GPU. The system has been flown end-to-end: a field trial produced 41 captures, 247 evidence artefacts and zero errors with the SoC thermally unthrottled throughout.

Index Terms—Unmanned aerial vehicles, search and rescue, thermal imaging, embedded computer vision, object detection, sensor registration, model quantisation.

I. INTRODUCTION

The first seventy-two hours after an earthquake, flood or landslide dominate survival statistics, and the scarce resource in that window is not willingness but *search coverage*. A ground team sweeps rubble slowly and at risk to itself. A small unmanned aircraft can sweep faster, but only if something aboard decides what deserves a human being's attention—streaming raw video to an operator simply relocates the bottleneck.

That decision has to be made onboard, which imposes a budget. The aircraft is battery-powered, so computation is flight time. The payload must cost a fraction of the airframe. And for the mission to mean anything at night, the primary sensor must be thermal.

The literature resolves this budget by spending money. The flight-tested system of Hinzmann *et al.* [1], developed with Switzerland's air-rescue service, pairs a FLIR Tau2 640×480 core with an NVIDIA Jetson TX2. Current multispectral detectors such as COXNet [3] and TFSANet [4] assume the same class of thermal imager and are trained and benchmarked on desktop GPUs. Their accuracy is not in question; their bill of materials is.

We took the opposite constraint as given: a 32×24 MLX90640 thermal array costing roughly 3% of a Tau2, a Raspberry Pi 4 with 1 GB of RAM, and a total payload budget near ₹14,000. This is not a smaller version of the same design. At a 20 m survey altitude the thermal ground sample distance is 651 mm, so a prone adult subtends **2.61 thermal pixels** (Section III-B). No amount of model capacity recovers a person from 2.6 pixels. The thermal path can therefore only ever be an *anomaly detector*, never a classifier of people—and once that is accepted, the architecture follows.

Our contribution is the arrangement rather than any single component:

- 1) A **non-learned thermal gate** built on robust statistics (median/MAD) that nominates candidate regions at essentially zero cost and is scale-free, so that identical code runs on radiometric degrees Celsius from the MLX90640 and on 8-bit public thermal datasets (Section IV-A).

- 2) A **cascade** in which the convolutional detector never sees a whole frame, reducing its work to between 1/16 and 1/3 of a whole-frame detector and making per-frame latency a function of scene content rather than image size (Sections IV-C, VI-F).

- 3) A **measured cross-modal registration** for a sensor pair whose 27 mm baseline and 45:1 resolution ratio make manual correspondence annotation infeasible, solved by automatic sub-pixel target detection (Section IV-D), together with an explicit depth-error analysis that fixes the calibration standoff distance (Section VI-D).

- 4) An **empirical result that contradicts the intuitive design**: the stride-4 P2 detection head, the standard prescription for small-object detection, was built, trained identically, and lost to P3–P5 in every object-size bin at twice the compute (Section VI-B).

- 5) A complete, individually validated **downstream stack**—ego-motion-compensated tracking, attitude estimation, georeferencing and ground-side track management—specified in Section IV-G to IV-J and inventoried with its deployment status in Table II, so that what flies today is distinguished from what is implemented and tested.

- 6) A **quantisation finding** with consequences beyond this system: full INT8 quantisation degraded the detector to 31.9% of reference accuracy *silently*—the model loaded, ran fast, and returned plausible boxes—because 8-bit activations corrupt the Distribution Focal Loss normalisation constant. Quantising weights

only (w8a32) recovers 99.2% at an identical file size (Section VI-C).

II. RELATED WORK

A. Flight-tested optical/infrared SAR

Hinzmann *et al.* [1] is the closest system-level reference and the most useful. They run two independent RetinaNet-50 detectors, one per modality, and—importantly for us—their fusion is *geometric rather than learned*: a thermal box is mapped into the optical frame by stereo calibration, a sliding window nine times the box size is searched over 36 positions, and matches at $\text{IoU} \geq 0.5$ are merged by a logical OR with averaged scores. Of all the fusion schemes surveyed here this is the only one transferable to our sensor pair, because it operates on *boxes* rather than feature maps. Their custom anchor scales raised training-contributing samples from 57.7% to 97% (optical) and 33.2% to 88.1% (thermal), solving by anchor design the same small-object problem we address in Section VI-B. Their headline metric—fewer than 30% of individuals missed across a mission at under one false positive per image—is deliberately a per-mission rather than per-frame quantity, which is structurally the same argument our log-odds ledger makes in Section IV-F.

B. Low-cost embedded SAR, and the result we set out to beat

De Oliveira and Wehrmeister [2] is the nearest match to our constraints in the whole literature: a Raspberry Pi 2, a FLIR Lepton 80×60 thermal imager, a Raspicam, and a 15–20 m survey altitude. Their measured result is the one we quote most often: **CNN inference with thermal on the Pi 2 ran at 0.18 FPS**—5.5 seconds per frame—against 14 FPS for HOG+SVM, leading them to conclude that CNNs "require many computational resources making real-time embedded processing difficult." They did not fuse the modalities; the two ran in parallel for different jobs. Their

classifier reached 99.71% on clean full pedestrians but 71.1% under 25% occlusion.

Our architecture is a direct answer to that conclusion. We reach 28.2 ms per crop on the same class of hardware not because the network is better but because the gate decides *where* to spend it. The comparison is developed in Section VII.

C. Learned RGBT fusion, and why it does not transfer

COXNet [3] fuses high-level visible with low-level thermal features through a wavelet-based alignment module, adds dynamic scale refinement and a geometry-aware label assignment, and reaches 47.18 mAP₅₀ on RGBTDronePerson at 51.27 GFLOPs and 71.11 M parameters; its P2–P6 variant reaches 51.82 at 123.59 GFLOPs. TFSANet [4] adds transformer fusion with scale-aware attention and reports 86.56 AP on NII-CU and +4.33 AP on RGBTDronePerson.

Both assume a 640×512 thermal input. Ours is 32×24—**768 pixels against 327,680, a factor of 426**. Pushing a 32×24 frame through a standard backbone yields 8×6 at stride 4 and 4×3 at stride 8: there is no thermal feature pyramid to fuse. This is an information limit, not a compute limit, and it is not recoverable by quantisation or by shrinking the model. We therefore adopt the geometric merge of [1] and cite [3], [4] as the state of the art we can characterise but not run.

D. Components

The detector is YOLOv8-nano [5], anchor-free with distributional box regression [7]. The scene classifier is MobileNetV2 [6] transferred to AIDER [10]. Robust scale estimation follows the MAD treatment of Rousseeuw and Croux [12]; the affine fit uses RANSAC [11]. Calibration quality is assessed with the expected-calibration-error framing of Guo *et al.* [15], and quantisation follows the weight/activation separation described by Jacob *et al.* [16].

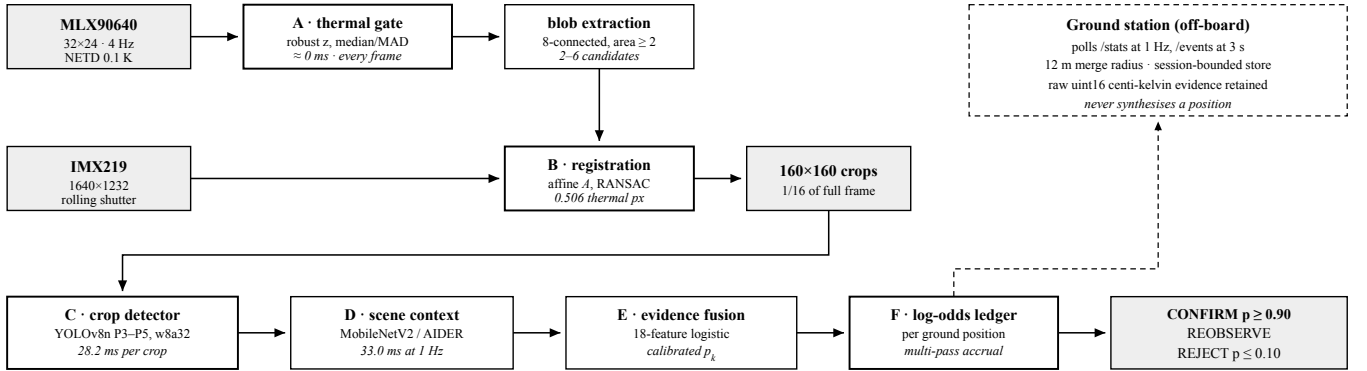


Fig. 1. The five-stage cascade, drawn in order of increasing cost. Stage A runs on every thermal frame and is not machine learning; stages C and beyond run only on what the stage before them passed. Heavy borders mark the three stages that consume measurable time. The dashed block is the off-board ground station, which polls the payload's HTTP endpoints rather than being polled—a deliberate choice, so that nothing new can crash in flight and a dropped radio link degrades to a stale timestamp rather than lost captures. Timings are medians measured on the target hardware (Section VI-F).

III. SYSTEM OVERVIEW

A. Hardware

The payload is assembled entirely from commodity parts (Table I). The compute element is a Raspberry Pi 4 Model B rev a03115 with 1 GB of RAM, of which roughly 905 MB is usable after the operating system. The runtime is deliberately minimal—ai-edge-litert 2.2.0, NumPy and headless OpenCV, with neither TensorFlow nor PyTorch installed—because the full frameworks would not fit alongside the models.

TABLE I. PAYLOAD BILL OF MATERIALS AND MEASURED PROPERTIES.

Element	Part	Property
Compute	Raspberry Pi 4B, 1 GB	905 MB usable
Thermal	Melexis MLX90640 [13]	32×24, 4 Hz
	field of view	55° × 35°
	noise-equivalent ΔT	0.1 K
Visible	Sony IMX219	1640×1232
	field of view	62.2° × 48.8°
	focal length	1359.3 px
Baseline	lens-centre separation	27 mm
IMU	MPU-6050	I ² C 0x68
GNSS	u-blox NEO-6M	NMEA, 9600 Bd
Bus	I ² C	400 kHz

The I²C rate is load-bearing rather than incidental: at the default 100 kHz the MLX90640 frame tears across its subpage boundary, producing a chequerboard artefact that survives into the gate as spurious contrast.

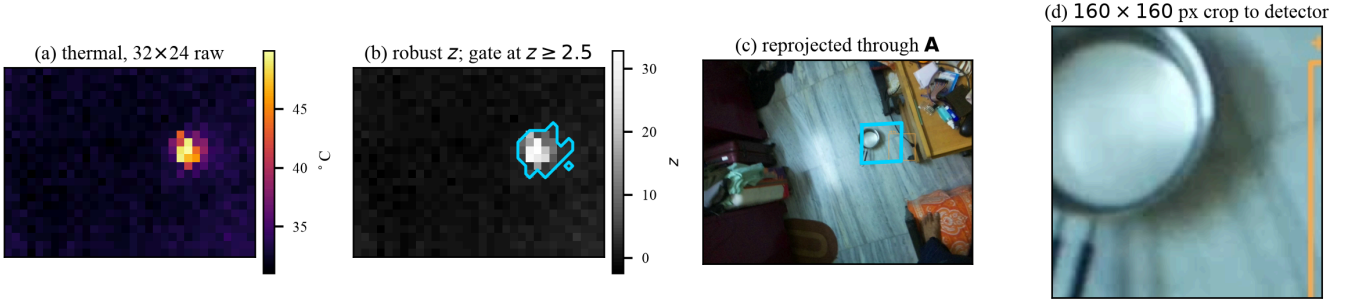


Fig. 2. The cascade on a single real capture from the payload (frame 215558, 2026-09-17). (a) the raw 32×24 radiometric frame in °C, rendered nearest-neighbour so that the pixel grid—the actual information available—is visible. Scene median $T_{bg} = 32.36$ °C. (b) the robust z field of (3); the gate contour at $z \geq 2.5$ encloses 31 pixels, with $\sigma = 0.535$ K and a peak of $z = 32.9$. Three further single-pixel components were discarded by the area test. (c) the blob's intensity-weighted centroid, at thermal coordinate (23.17, 10.06), reprojected through the fitted affine $\mathbf{A}$ into the visible frame at (1058, 504)—landing on the target. (d) the 160×160 crop actually handed to the detector, 1/64 of the visible frame area. The faint orange rectangle in (c) and (d) is the payload's own on-screen overlay from an earlier frame and is not part of this measurement.

IV. METHOD

A. Stage A: robust thermal gating

The gate answers one question per frame: which pixels differ from this scene's own background more than sensor noise and clutter explain? It is a statistical test, not a learned model, and that is deliberate—it must run on every frame within a budget of approximately zero.

The obvious formulation, $z = (t - \mu)/s$ with sample mean and standard deviation, fails for a specific reason: a warm body in the frame raises both μ and s , inflating the very threshold intended to detect it. The breakdown point of the standard deviation is zero. We therefore estimate background and scale with the median and the median absolute deviation [12], whose breakdown point is 50%:

$$T_{bg} = \text{med}(T), \quad \text{MAD} = \text{med}(|T - T_{bg}|) \quad (2)$$

$$\sigma = \max(1.4826 \cdot \text{MAD}, \sigma_{\min}), \quad z = \frac{T - T_{bg}}{\sigma} \quad (3)$$

The factor 1.4826 makes σ a consistent estimator of the standard deviation under Gaussian noise, so that the threshold retains its familiar interpretation. The floor $\sigma_{\min} = 0.15$ K prevents division by a collapsed scale in a perfectly flat scene, where MAD tends to

B. What 32×24 pixels buys

The design follows from a single arithmetic fact. For a sensor of horizontal field of view θ and N pixels across, at altitude h , the ground sample distance is

$$\text{GSD} = \frac{2h \tan(\theta/2)}{N}. \quad (1)$$

At $h = 20$ m with $\theta = 55^\circ$ and $N = 32$, the thermal swath is 20.82 m wide and GSD = 651 mm. A prone adult of 1.7 m therefore spans **2.61 thermal pixels**. The visible camera at the same altitude gives GSD = 14.7 mm, so the same person spans about 116 pixels.

Two consequences shape everything downstream. First, the thermal sensor cannot classify: 2.6 pixels carry the presence of an anomaly, its rough extent and its sign, and nothing else. Second, the visible camera *can* classify but cannot be searched exhaustively within the power budget. The cascade exists to route the thermal sensor's cheap, high-recall evidence into the expensive, discriminative one.

zero and any quantisation step would otherwise read as infinite contrast. A pixel is nominated when

$$|z| \geq z_t = 2.5 \quad \wedge \quad |T - T_{bg}| \geq \Delta T_{\min} = 0.5 \text{ K}. \quad (4)$$

Two details in (4) are mission-specific. The test is *two-sided*: on sun-heated concrete or rubble a human body is frequently *colder* than its surroundings, and a one-sided "warmer than background" test—the intuitive choice—misses exactly that regime. The additive $\Delta T_{\min}$ term, expressed in kelvin rather than in units of σ , stops the gate firing on pure sensor noise when σ has hit its floor. Because z is dimensionless, the same thresholds apply unchanged to radiometric Celsius and to the 8-bit thermal imagery of public datasets, which is what allows the gate to be evaluated offline at all.

Nominated pixels are grouped into 8-connected components; components below 2 pixels are discarded as speckle. For each surviving blob we record the peak z , the signed peak ΔT , the sign, the area in thermal pixels, and the eccentricity of its second-moment ellipse. The blob's position is taken as the *intensity-weighted* centroid

$$\mathbf{c} = \frac{\sum_{i \in B} z_i \mathbf{x}_i}{\sum_{i \in B} z_i}, \quad (5)$$

which recovers sub-pixel position from a handful of pixels and is the reason the registration of Section IV-D can be fitted at all.

B. Stage B: candidate crops

Each blob centroid is mapped into visible-image coordinates by (6) and a square crop is cut around it—160 px for typical blobs, 224 px when the blob's reprojected extent demands it. The gate is tuned permissively, yielding two to six crops per frame; in search and rescue a false alarm costs an operator a few seconds while a miss can cost a life, so the burden of rejection is deliberately pushed onto the later, more discriminative stages.

C. Stage C: the crop detector

The detector is a YOLOv8-nano [5] with the P3–P5 head (strides 8, 16, 32), 3,011,628 parameters at four classes and 8.19 GFLOPs at 640 px. At the deployed 160 px crop size the head exposes 525 anchor points against 8,400 at 640 px.

YOLOv8 is anchor-free and regresses each box side as a distribution rather than a scalar [7]. Each side d is represented by a softmax over $r_{\max} = 16$ bins and decoded as the expectation

$$d = \sum_{j=0}^{15} j \cdot \text{softmax}(\mathbf{I})_j, \quad (6)$$

which lets the network express genuine ambiguity about an edge—valuable when the object is twelve pixels tall. Equation (6) is also precisely where quantisation breaks the model (Section VI-C). Non-maximum suppression runs in NumPy on the host rather than inside the graph, avoiding a second framework dependency.

The choice of the nano scaling is load-bearing rather than a default. By the time a crop reaches the detector, the gate has already asserted that something anomalous is centred in it; the detector answers a much easier question than a whole-frame detector does.

D. Cross-modal registration

Mapping a thermal blob into the visible frame requires a transform between sensors whose resolutions differ by 45:1—one thermal pixel subtends about 45 visible pixels. We model the mapping as a six-parameter affine

$$\mathbf{x}_{\text{rgb}} = \mathbf{A} [u \ v \ 1]^T, \quad \mathbf{A} \in \mathbb{R}^{2 \times 3}, \quad (7)$$

fitted by RANSAC least squares [11] over N correspondences, minimising

$$\varepsilon = \frac{1}{N} \sum_i \|\mathbf{A} \mathbf{u}_i - \mathbf{x}_i\|_2. \quad (8)$$

An affine rather than a homography is appropriate here because the thermal array's angular resolution cannot resolve the perspective terms; adding them fits noise.

The practical obstacle is annotation. A human clicking a correspondence to the nearest thermal pixel commits, after scaling, a 45-pixel error in the visible frame—larger than any sensible RANSAC inlier threshold, so hand-labelled points are not merely imprecise, they are systematically rejected. We therefore detect correspondences automatically: the thermal target is localised to sub-pixel accuracy by (5), and its visible counterpart is found by a Hough circle search restricted to a window predicted from the thermal centroid through the nominal field-of-view ratio. A heated vessel of water served as the target, giving both a strong thermal signature and a circular visible outline. Thirty-eight correspondences were collected over a spatial spread selected by greedy farthest-point sampling, with zero operator clicks.

Because an affine is exact for a single plane at a single depth, the calibration standoff is not arbitrary. For a stereo pair of baseline b and focal length f , a calibration at range Z_c applied at operating range Z_f leaves a residual disparity

$$\delta = fb \left| \frac{1}{Z_c} - \frac{1}{Z_f} \right| \text{ [visible px]}. \quad (9)$$

Equation (9) is what fixes the bench procedure, and Section VI-D reports its consequences.

E. Stage D: scene context

A MobileNetV2 [6] classifier, transferred from ImageNet and fine-tuned on AIDER [10], labels the visible frame as one of *collapsed building*, *fire*, *flooded area*, *traffic incident* or *normal*. It runs on its own thread at 1 Hz—scene type changes on the scale of a flight leg, not a frame—so its 33.0 ms cost amortises to 8.25 ms per thermal frame. The thread is fail-safe: if the classifier throws or is absent the cascade continues with the context features zeroed, because a scene label is contextual evidence and never a gating condition.

F. Stage E–F: evidence fusion and the multi-pass ledger

Each candidate is summarised by an 18-element feature vector spanning five evidence sources: thermal shape (z_{peak} , ΔT , sign, area, eccentricity), the visible detector (score, presence flag, relative box area), registration agreement (IoU, normalised centroid distance), temporal persistence (hits, hit ratio), and scene context (T_{bg} , luminance, the three hazard probabilities, altitude band). A logistic regression maps this to a calibrated probability

$$p_k = \sigma(\mathbf{w}^T \mathbf{f} + b), \quad \sigma(x) = (1 + e^{-x})^{-1}. \quad (10)$$

A linear model is chosen over a stronger one because calibration matters more than ranking here: the ledger below *adds* these quantities across passes, so a systematic bias compounds rather than averaging out, and an operator must be able to read the number as a probability.

A ground position observed on several passes accumulates evidence in log-odds. Writing π_0 for the prior, the ledger for a target after K passes is

$$L = \text{logit}(\pi_0) + \sum_{k=1}^K w_k [\text{logit}(p_k) - \text{logit}(\pi_0)], \quad (11)$$

$$w_k = 1 \text{ if the altitude band is new, } 0.7 \text{ otherwise.} \quad (12)$$

The discount in (12) is the one departure from naive Bayesian accrual, and it corrects a real failure mode: repeated observations from the same altitude and aspect are strongly correlated, so treating them as independent would drive a persistent thermal artefact—a sun-warmed rock—to certainty simply by looking at it repeatedly. Requiring a new viewing geometry to earn full weight makes the ledger reward genuinely new evidence. The final probability is $p = \sigma(L)$, and the decision is CONFIRM at $p \geq 0.90$, REJECT at $p \leq 0.10$, and otherwise re-observation at a lower altitude band until a pass limit is reached. This mission-level rather than frame-level criterion mirrors the evaluation protocol of [1].

G. Temporal association: ego-motion compensation and tracking

A survey aircraft's own motion dominates a survivor's. At 14 m/s and 20 m with $f = 1359.3$ px, a *stationary* ground point translates $f v \Delta t/h = 95$ px per 100 ms in the visible image. A constant-velocity tracker initialised from one detection with zero velocity therefore loses a stationary casualty on the very next frame. Association must run on ego-motion-compensated predictions.

We predict the inter-frame image motion from sensors already carried, rather than estimating it from the images. Given GNSS ground speed v and course χ , camera yaw ψ , and body rates (p, q, r) from the gyroscope, the frame-to-frame map is a similarity transform with

$$t_x = \frac{fv\Delta t}{h} \sin(\chi - \psi) + fq\Delta t, \quad \theta = r\Delta t, \quad (13)$$

with t_y taking the cosine term and $fp\Delta t$. A track's predicted position is then rotated about the image centre and translated, $\mathbf{x}' = \mathbf{R}(\theta)(\mathbf{x} - \mathbf{c}) + \mathbf{c} + \mathbf{t}$, before the motion filter runs. This ordering is the whole point: the filter is then left modelling only the target's own motion, which for a survivor is zero.

Residual motion is carried by a four-state constant-velocity Kalman filter [21] with state $\mathbf{x} = (u, v, \dot{u}, \dot{v})$ in image pixels:

$$\mathbf{x}^- = \mathbf{F}\mathbf{x}, \quad \mathbf{P}^- = \mathbf{F}\mathbf{P}\mathbf{F}^T + \mathbf{Q} \quad (14)$$

$$\mathbf{K} = \mathbf{P}^-\mathbf{H}^T(\mathbf{H}\mathbf{P}^-\mathbf{H}^T + \mathbf{R})^{-1}, \quad \mathbf{x} = \mathbf{x}^- + \mathbf{K}(\mathbf{z} - \mathbf{H}\mathbf{x}^-) \quad (15)$$

with $\mathbf{Q} = 4\mathbf{I}$, $\mathbf{R} = 9\mathbf{I}$ px² and $\mathbf{H}$ selecting position. Detections are assigned to tracks by the Hungarian algorithm [17] on the cost

$$C_{ij} = 1 - \text{IoU}(b_p, d_j), \quad \text{accepted only if } \text{IoU} \geq 0.3, \quad (16)$$

following SORT [18]. A track is confirmed on 2 hits within a 3-frame window and deleted after 5 consecutive misses. The same predicted transform, rescaled into thermal-pixel units by the measured across- and along-track ratios (45.3 and 36.8 visible px per thermal px), drives a blob persistence counter in the 32×24 thermal frame with a 1.5 thermal-pixel match radius; that counter supplies the `hits` and `hit_ratio` features of (10).

H. Georeferencing a detection

A pixel becomes a map coordinate through the platform's attitude. Roll and pitch are recovered from the accelerometer's gravity direction, $\phi = \text{atan2}(a_y, a_z)$ and $\theta = \text{atan2}(-a_x, \sqrt{a_y^2 + a_z^2})$, and fused with the gyroscope by a complementary filter

$$\theta_k = \alpha(\theta_{k-1} + \omega\Delta t) + (1-\alpha)\theta_{\text{acc}}, \quad \alpha = 0.98, \quad (17)$$

after a start-up gyro-bias average over stationary samples. The filter is a deliberate choice over a full attitude Kalman filter: with one accelerometer and one gyroscope the steady-state solution is a fixed blend, and (17) simply starts there. Yaw is not observable from this sensor—there is no magnetometer—so heading comes from GNSS course over ground while the aircraft is moving, which is stated here because it is a real constraint on the localisation claim rather than an implementation detail.

The pixel is then cast as a ray and intersected with a level ground plane:

$$\mathbf{d}_g = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi) \left[\frac{u-c_u}{f}, \frac{v-c_v}{f}, 1 \right]^T, \quad s = h/d_{g,z} \quad (18)$$

giving east and north offsets $s d_{g,x}$ and $s d_{g,y}$, converted to latitude and longitude at 111,320 m per degree with the usual $\cos(\text{lat})$ scaling on longitude. The flat-ground assumption and the GNSS fix dominate this error budget; the rotation convention does not.

I. Ground-station track management

Confirmed detections arriving at the ground station are maintained as map tracks by a two-axis alpha-beta filter in local metres rather than degrees—filtering in degrees would make the east and north gains differ by $\cos(\text{lat})$, a silent bug that appears as a track drifting sideways. Gains are not hand-tuned but taken from Kalata's steady-state solution [19] for the tracking index $L = \sigma_a T^2 / \sigma_z$:

$$r = \frac{4 + L - \sqrt{(8L + L^2)}}{4}, \quad \alpha = 1 - r^2, \quad \beta = 2(1 - r)^2. \quad (19)$$

This leaves exactly one physical knob, the ratio of how far process noise can move the target in one scan to how well it can be

measured, and it costs two multiplies per axis—which matters because the same code must run on the Pi. Benedict–Bordner variance reduction factors [20] turn α, β into smoothed position and velocity variances, so the uncertainty ellipse drawn for an operator carries a computed number rather than a guess. A plot is associated only inside a validation gate of radius

$$R_{\text{gate}} = \sqrt{(\chi^2_{0.9973, 2} (\sigma_{\text{pred}}^2 + \sigma_{\text{meas}}^2))}, \quad \chi^2_{0.9973, 2} = 11.83. \quad (20)$$

The 99.73% point is chosen deliberately: in search and rescue, splitting one survivor across two tracks costs far less than silently discarding a real detection. Tracks are initiated M-of-N (3 of 4) and move through TENTATIVE, FIRM, COAST and either DROPPED or HELD after five consecutive coasts.

One profile deserves explicit mention. For a casualty the filter runs in a *static* mode with $\alpha = 1, \beta = 0$ and position variance fixed at σ_z^2 . A survivor lying in rubble is not moving, so the constant-velocity dead-reckoning term $\frac{1}{2}at^2$ is not merely unhelpful but actively misleading—left enabled it draws a 300 m uncertainty circle around someone who has not moved a metre. A fix on a stationary object does not become less certain with time; it becomes *older*, and age is displayed separately.

J. Verdict assignment

The final label is assigned from the evidence *pattern*, not from p alone. A high score with no visible corroboration is capped at MEDIUM, and a visible detection with no thermal support ($z_{\text{peak}} < z_t$) is forced to LOW regardless of its own score, because in this system a person is a thermal object first. HIGH requires $p \geq 0.80$ together with visible corroboration at $\text{IoU} \geq 0.10$ and at least two hits.

TABLE II. COMPLETE INVENTORY OF MODELS AND ALGORITHMS.

Component	Method	Status
Thermal gate	robust z , median/MAD (2)–(4)	payload
Blob extraction	8-connected components, weighted centroid (5)	payload
Registration	6-parameter affine, RANSAC (7), (8)	payload
Person detector	YOLOv8n P3–P5, w8a32, 3.01 M param	payload
Box decode	DFL expectation (6) + greedy per-class NMS, NumPy	payload
Visible detector	YOLOv8n COCO-80, w8a32	payload
Scene classifier	MobileNetV2, INT8, 2.74 MB	payload
Ego-motion	GNSS+gyro similarity transform (13)	tested
Motion filter	4-state constant-velocity Kalman (14), (15)	tested
Association	SORT, Hungarian on 1–IoU (16)	tested
Blob persistence	ego-compensated, thermal-space, 1.5 px	tested
Attitude	complementary filter, $\alpha = 0.98$ (17)	tested
Georeferencing	pinhole ray / ground-plane intersection (18)	tested
Evidence fusion	logistic regression, 18 features (10)	instrumented
Confidence ledger	log-odds, correlation-discounted (11), (12)	tested
Verdict rules	pattern-based three-class assignment	tested
Map tracker	alpha-beta, Kalata gains (19), χ^2 gate (20), 3-of-4	ground

payload: runs in the flying build on the Pi. **ground:** runs in the ground station. **tested:** implemented and unit-tested against the acceptance criteria quoted in the text, but not in the current flying build—the payload build is deliberately minimal, and the tracking and georeferencing stages require the GNSS/IMU stream that indoor bench trials cannot supply. **instrumented:** trained and evaluated, reported in Section VIII-B, and not relied upon by any claim in this paper. The suite is 180 tests.

V. DATASETS AND TRAINING

The detector is trained on the union of two independently collected aerial-thermal corpora. HIT-UAV [8] provides clean, largely overhead campus scenes with four annotated classes. RGBTDronePerson [9] provides what HIT-UAV lacks—urban, night, oblique drone views—and annotates people only. Training across both answers a question one dataset cannot: whether the detector has learned *people* or has learned *HIT-UAV*.

TABLE III. DATASET COMPOSITION AFTER CONVERSION.

Corpus	train	val	classes
HIT-UAV [8]	4,058	580	4
RGBTDronePerson-thermal [9]	4,900	1,225	1
<i>combined</i> (2,029 + 4,900)	6,929	1,515	4
AIDER [10] (scene)	—	1,286	5

HIT-UAV is subsampled into the combined set to limit its dominance. Files are hard-linked rather than copied, and carry `hit_`/`rgbt_` filename prefixes precisely so that the validation split can be scored per source.

Two protocol decisions deserve statement. First, the label spaces are merged at four classes rather than reduced to one: class 0 is *Person* in both corpora, so keeping `nc=4` carries the pretrained detection head over intact, whereas dropping to `nc=1` would rebuild the class branch for no benefit. The accepted cost is that RGBTDronePerson frames contain unlabelled cars and bicycles, which become false negatives and degrade those classes. This is acceptable for a payload that acts only on *Person*, and is the reason we report Person AP_{50} throughout and never overall mAP.

Second, **evaluation is per source, never pooled**. The combined validation split is 81% RGBTDronePerson by count, so a single pooled number principally reports performance on the harder corpus while appearing to describe both. A combined model is treated as an improvement only if it holds HIT-UAV Person AP while raising RGBT recall—a rule we apply against our own candidate model in Section VI-A.

Training starts from COCO-pretrained weights on Apple-silicon MPS at 640 px, batch 12, with mosaic augmentation disabled for the final epochs. Checkpoint selection uses the Ultralytics fitness $0.1 \text{ mAP}_{50} + 0.9 \text{ mAP}_{50-95}$, so the retained checkpoint is chosen by the strict localisation metric rather than the headline one. A watchdog monitors the loss trace and aborts on non-finite or negative loss, on a threefold loss explosion, or on three consecutive epochs below half the best observed mAP_{50} —a guard added after an fp16 failure produced a classification loss running $4.97 \rightarrow 46 \rightarrow -1224$ without raising an exception.

VI. EXPERIMENTS AND RESULTS

A. Detector accuracy, per source

Table IV reports the shipped detector against each corpus separately. The 0.851 HIT-UAV figure is directly comparable to the

single-corpus baseline, and the 0.478 RGBT figure shows what generalisation to a genuinely different aerial-thermal domain currently costs.

TABLE IV. PERSON DETECTION, EVALUATED PER SOURCE.

Model / corpus	imgs	AP_{50}	AP_{50-95}
Shipped (45 ep) · HIT-UAV	580	0.851	0.379
Shipped (45 ep) · RGBT	1,225	0.478	0.153
<i>mean of the two</i>	—	0.664	—
Continued (85 ep) · HIT-UAV	290	0.868	0.419
Continued (85 ep) · RGBT	1,225	0.431	0.128
<i>mean of the two</i>	—	0.649	—

The 85-epoch continuation gains 0.017 on HIT-UAV and loses 0.047 on RGBT. Mean Person AP_{50} falls, so it fails the adoption rule of Section V and was **not** deployed. Its value was diagnostic: performance is flat from epoch 25 through 40 across a mosaic-augmentation transition, which indicates the training-duration lever is exhausted.

This negative result is worth stating plainly because a pooled score would have concealed it. Pooled mAP_{50} rose from 0.668 to 0.683 over the continuation—an apparent improvement that was in fact a trade of the harder domain for the easier one.

B. Detection head: P2–P5 against P3–P5

Small-object detection conventionally prescribes adding a stride-4 (P2) feature level. We built it, trained it identically, and it lost. Table V and Fig. 3 give the comparison.

TABLE V. HEAD ABLATION, HIT-UAV, 30 EPOCHS, IDENTICAL SCHEDULE.

Head	anchors	GFLOPs	Person AP_{50}	mAP_{50}
P2–P5 (strides 4–32)	34,000	17.43	0.801	0.588
P3–P5 (strides 8–32)	8,400	8.86	0.821	0.639

Anchor counts at 640 px. P2 costs $1.97\times$ the arithmetic and $4.0\times$ the anchor points for lower accuracy in every size bin.

The explanation is that P2's benefit is conditional on the small object being *resolved but coarsely sampled*. In our regime the limiting factor is information rather than sampling density, and the extra level mainly adds anchors that must be suppressed. Fig. 3(b) makes the point sharply: as images are progressively downscaled—simulating higher altitude—both heads collapse together, and neither crosses 0.20 recall once 8–16 px objects are reduced to a quarter scale. The system is information-limited, not capacity-limited. The same reasoning is why we did not migrate to a newer backbone: a candidate YOLO11n at four classes carries 2,590,620 parameters against our 3,011,628, a 14.0% reduction in capacity that addresses a constraint which is not binding.

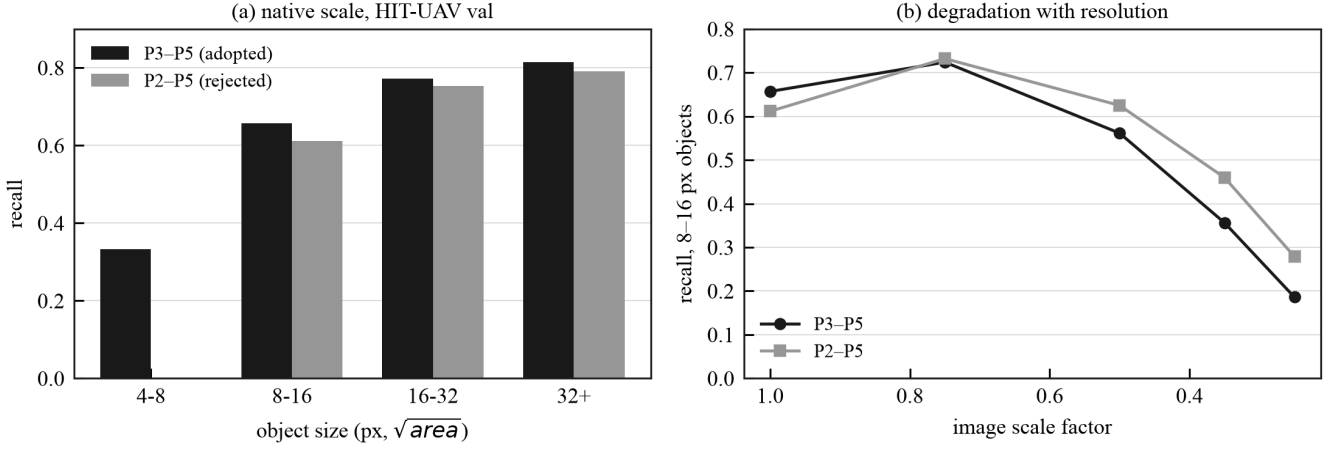


Fig. 3. Recall as a function of pixels on target, HIT-UAV validation. **(a)** at native scale the P3–P5 head matches or beats P2–P5 in every size bin while using half the arithmetic; the advantage is largest exactly where P2 was expected to help, in the 8–16 px bin (0.657 against 0.612). **(b)** recall on 8–16 px objects as the image is downsampled, simulating increasing altitude. Both heads fall together, which identifies the binding constraint as available information rather than model capacity.

C. Quantisation: a silent failure and its cause

Deployment requires an 8-bit export. Full INT8—weights *and* activations—is the default choice and was catastrophic here: the exported model scored **31.9% of the reference model's accuracy**. Critically, it failed silently. It loaded, ran at the expected speed, and returned plausible-looking boxes; only a quantitative comparison against the float reference exposed it.

The cause is (6). The DFL decode is a weighted sum over 16 softmax bins followed by a normalisation by the input size. When activations are quantised, that normalisation constant is quantised with them: the effective divisor measured on the failing export was ≈ 636 instead of 640. A 0.6% scale error shifts every box centre by roughly 2.5 px—enough to drop IoU below threshold for small objects while leaving the output visually reasonable.

Quantising weights only (w8a32) fixes it at *identical file size*, because weights dominate storage while activations dominate arithmetic. Table VI reports the adopted export against its float reference.

TABLE VI. W8A32 EXPORT FIDELITY, 290 IMAGES, 1,168 ANNOTATED PERSONS.

Quantity	value
Person AP ₅₀ , float32 reference	0.8499
Person AP ₅₀ , w8a32 export	0.8599
absolute difference	+0.0100
box-row correlation	0.99972
class-row correlation	0.99812
box-row MAE	2.24 px
file size (160 / 224 / 640 px)	3.23 / 3.23 / 3.32 MB

The export scores marginally *above* its float reference; at this sample size the difference is not meaningful, and the row correlations are the substantive evidence of fidelity. Full INT8 on the same comparison scored 31.9% of reference.

D. Registration accuracy and the calibration standoff

The affine of (7) was fitted over 38 automatically-detected correspondences, giving

$$\mathbf{A} = \begin{bmatrix} 34.466 & -0.452 & 263.99 \\ -1.758 & 32.881 & 214.31 \end{bmatrix} \quad (21)$$

The mean reprojection residual is 22.76 visible pixels, which at the nominal 45 visible pixels per thermal pixel is **0.506 thermal pixels**—against 3.74 thermal pixels for the nominal field-of-view scaling

it replaces, a **7.4×** improvement. The diagonal terms of (13) recover the 45:1 scale to within 3%, and the small off-diagonal terms quantify a residual sensor-mounting rotation of roughly 1.5° that the fit absorbs and a ratio-scaling approach cannot.

Equation (9) determines where the calibration target must stand. With $b = 27$ mm and $f = 1359.3$ px, calibrating at 1 m and flying at 20 m leaves 0.775 thermal pixels of depth-induced error—which alone exceeds the 0.5 px acceptance threshold, so a bench-distance calibration cannot pass no matter how carefully the correspondences are measured. Calibrating at 2 m reduces this to 0.367 px and passes. Fig. 4 plots the relation; the practical rule is that the calibration target must stand at least 2 m from the sensor pair.

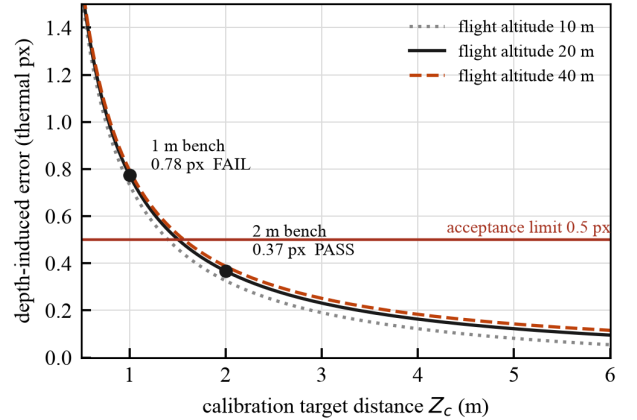


Fig. 4. Depth-induced reprojection error from (9) for the measured 27 mm baseline. The acceptance limit is 0.5 thermal px. A 1 m bench calibration cannot meet it at any operating altitude; 2 m can. The curve flattens beyond about 3 m, so there is no benefit in a longer and more awkward standoff.



Fig. 5. Registration verification: the thermal field warped through **A** and composited onto four visible frames spanning different target positions in the field of view. The hot region tracks the vessel across the frame, including near the edges where a pure scale-and-offset mapping diverges most.

E. Scene classification

The MobileNetV2 classifier attains **95.2% accuracy** over 1,286 validation images with a macro F1 of 0.927, exporting to 2.74 MB in INT8. AIDER is 68% *normal*, so training is class-weighted and the macro average is the honest summary. Per-class recall is 0.979 (collapsed building), 0.981 (fire), 0.958 (flooded areas), 0.915 (traffic incident) and 0.949 (normal).

The error structure in Fig. 6 is the useful part: essentially all confusion is *normal* scenes predicted as a hazard—21 as collapse, 10 as flood, 9 as traffic, 6 as fire—while hazard scenes are rarely predicted normal. For a search-and-rescue payload this asymmetry is the correct direction of error, and it is reinforced by the architecture: the scene label is a contextual feature entering (10), never a gate on whether a candidate is examined.

	collapse	fire	flood	traffic	normal
collapse	94	1			1
fire		105			2
flood			91	1	3
traffic	3		1	86	4
normal	21	6	10	9	848
	collapse	fire	flood	traffic	normal

Fig. 6. Scene classifier confusion matrix, 1,286 validation images, shaded by row-normalised rate and annotated with counts. Off-diagonal mass lies almost entirely in the bottom row: the model over-calls hazards on normal scenes rather than missing hazards.

F. On-device latency and the frame budget

All timings in Table VII are medians measured on the flight hardware, not extrapolated. The thermal sensor runs at 4 Hz, giving a 250 ms budget per cycle.

TABLE VII. MEASURED LATENCY, RASPBERRY PI 4B 1 GB, DEBIAN 13.

Operation	median	p90
Thermal gate, 32×24	≈ 0 ms	—
Detector, 160 px crop	28.2 ms	28.3 ms
Detector, 224 px crop	56.7 ms	56.9 ms
Scene classifier, 640×480	33.0 ms	33.1 ms
Peak resident memory, all models	95.2 MB	—
SoC temperature under load	62–66.7 °C	—

The p90 within 0.4% of the median is itself a result: with the model resident and the tensor arena allocated once at load, per-frame latency has almost no tail, which is what makes the budget below a guarantee rather than an average.

The worst case is six crops at 160 px (169.2 ms) plus the scene classifier amortised over four frames (8.25 ms), totalling **177.5 ms against the 250 ms budget—a 29% margin** (Fig. 7). The typical two-crop case consumes 64.7 ms. Memory is the comfortable resource: 95.2 MB peak of 905 MB available, with an 8.39 MB tensor arena allocated once rather than per frame.

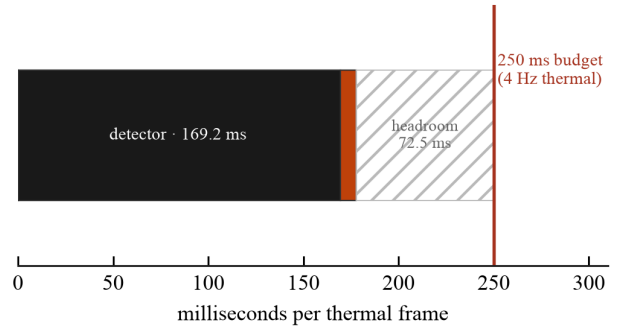


Fig. 7. Worst-case frame budget. The detector dominates; the gate is free and the scene classifier amortises. Because cost scales with the number of gate nominations rather than with image size, latency is a function of scene content and stays bounded.

G. Visible-branch confirmation

The visible branch runs a stock COCO YOLOv8n for daylight confirmation. Evaluated over 537 daytime frames containing 13,422 annotated person instances (11 night frames excluded), overall recall is 0.305, but the aggregate is misleading—recall is a steep function of apparent size (Fig. 8), rising from 0.075 below 16 px to 0.934 above 72 px. The median instance in this corpus is 16.9 px, which is why the pooled figure is low. At the 20 m design altitude the visible GSD of 14.7 mm places a standing adult near 116 px, in the regime where the branch is reliable. The measurement establishes the operating envelope rather than a deficiency: visible confirmation is trustworthy above roughly 32 px of apparent target size and should not be relied on below it.

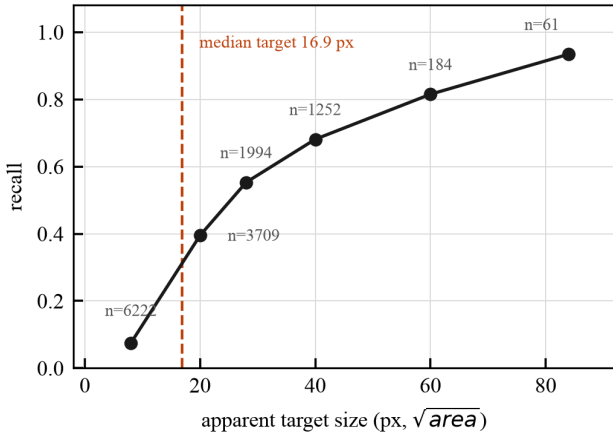


Fig. 8. Visible-branch recall against apparent target size, 13,422 instances. The dashed line marks the median instance size in this corpus. Recall is governed by pixels on target, the same quantity that governs the thermal branch.

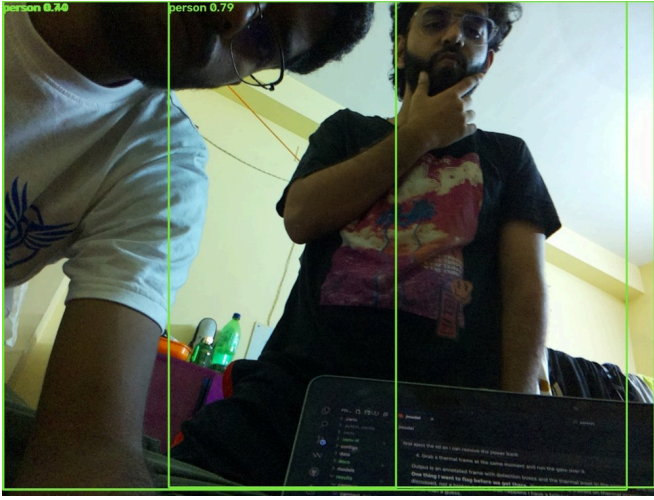


Fig. 9. Visible-branch output at close range from the payload's own camera, with detection confidences. At this apparent size the branch operates in the high-recall regime of Fig. 8.

H. End-to-end field trial

The complete system was flown as an integrated payload on 17 September 2026: thermal gate, calibrated reprojection, crop detection, scene classification at 1 Hz, live contour rendering through **A**, and telemetry to the ground station. The session produced **41 captures resolving to 1 target** and 247 evidence artefacts with **zero errors**. The SoC held 62–66.7 °C with the throttle register at 0×0 throughout—no thermal or undervoltage throttling occurred—the thermal loop held 4.0 Hz, and the storage device reported 25 GB free and no I/O errors.

The 41-to-1 ratio is itself the result of a design correction. Captures are merged into a single target within a 12 m radius; before this was applied to captures lacking a position fix, one stationary person generated 25 separate "targets". Unlocated captures now fold into the most recent unlocated target, with a 90 s gap starting a new one. Evidence is archived as raw uint16 centikelvin arrays (1,536 B per frame) rather than colour-mapped images, so a reviewer receives temperatures rather than a screenshot of a palette.

I. Ego-motion compensation: ablation

The tracking stage was validated against the worst case of Section IV-G: a stationary target observed for 50 frames at 14 m/s and 20 m altitude, $\Delta t = 100$ ms, which drives a 95.15 px per frame

apparent drift against a 34 px detection box. The two arms differ only in whether (13) is applied before the filter predicts.

TABLE VIII. EGO-MOTION ABLATION, 50 FRAMES, STATIONARY TARGET.

Arm	tracks created	hits	confirmed
Compensated (13)	1	50	yes
Uncompensated	> 10	—	none

With a 95 px inter-frame drift and a 34 px box, IoU between a track's prediction and the next detection is zero after the first frame, so the uncompensated tracker can never re-associate and spawns a fresh tentative track almost every frame—the failure is total rather than gradual.

This is the result that justifies predicting image motion from GNSS and gyroscope rather than estimating it from the images: the compensation is not a refinement that improves track quality, it is the precondition for any track existing at all at survey speed.

VII. COMPARISON WITH THE STATE OF THE ART

Table IX places the system against the reference works. The rows are not commensurable as accuracy comparisons—different sensors, corpora and tasks—and are not offered as such. They compare *what a given accuracy costs*.

TABLE IX. COST OF INFERENCE ACROSS COMPARABLE SYSTEMS.

System	params	GFLOPs	compute
QFDet* [3]	60.3 M	242.8	RTX 2080Ti
COXNet [3]	71.1 M	51.3	RTX 2080Ti
COXNet* [3]	75.9 M	123.6	RTX 2080Ti
ETH RetinaNet-50 ×2 [1]	~2×34 M	—	Jetson TX2
Sensors 2018 CNN [2]	—	—	Pi 2, 0.18 FPS
SaResQ (this work)	3.01 M	8.19	Pi 4 CPU

GFLOPs at the authors' stated input sizes. Our figure is for a whole 640 px frame; the deployed crop path costs a sixteenth of it.

Against [2]—the closest comparison, being thermal-assisted CNN detection on a Raspberry Pi at a 15–20 m survey altitude—the difference is two orders of magnitude in throughput. Their measured 0.18 FPS corresponds to 5,556 ms per frame; our worst case is 177.5 ms, a **31×** improvement before accounting for the generational difference between a Pi 2 and a Pi 4. Their conclusion that CNNs "require many computational resources making real-time embedded processing difficult" was correct for the architecture they evaluated—whole-frame inference—and is the premise our cascade is designed to falsify.

Against [3] and [4] the comparison is one of feasibility rather than quality. COXNet at 71.1 M parameters is 23.6× our size and needs a discrete GPU for 12.9 FPS; on a Pi 4 CPU the same graph would take tens of seconds per frame. More fundamentally, both methods fuse thermal *feature maps*, and our 32×24 input admits no such pyramid. We adopt instead the geometric box-level merge of [1], which is the one fusion mechanism in this literature that survives the resolution gap.

VIII. DISCUSSION

A. What the results establish

The load-bearing claim is that ordering computation by cost, and letting a nearly free statistical test decide where the expensive model looks, converts an infeasible workload into a feasible one without changing the model. Every measurement here supports that: the detector is unremarkable, and its 28.2 ms is unremarkable; what makes the system work is that it runs 2–6 times on 1/64 of the frame area rather than once on all of it.

A secondary theme is that two design decisions which appeared obvious were wrong and were caught only by measurement—the P2 head (Section VI-B) and full INT8 quantisation (Section VI-C). The second is the more dangerous, because it degraded accuracy by 68% while leaving every observable symptom normal.

B. Limitations

The fusion head of (10) is currently weak: ROC AUC 0.670, average precision 0.112, Brier 0.0243 against a base-rate baseline of 0.0247. Its expected calibration error is low (0.0058), so the probabilities it emits are honest, but its ranking power is limited—a direct consequence of nine of eighteen weights being exactly zero, because the registration-agreement features (IoU, centroid distance) and the track-persistence features are not measurable from a still-image training corpus. The registration of Section IV-D now makes the first group computable, and flight data will supply the second. We report the fusion head as instrumented but not yet validated, and no claim in this paper depends on it.

Gate recall has not been measured in the operating domain. The available offline figure, 2.9% on RGBTDronePerson, evaluates a contrast gate tuned for sparse targets over rubble against dense urban night scenes containing roughly ten people per frame; it is a domain mismatch rather than a performance estimate, and we decline to derive a sweep width from it. Measuring gate recall on MLX90640 imagery over representative terrain is the principal outstanding validation.

The tracking and georeferencing stages of Sections IV-G to IV-I are implemented and unit-tested against their acceptance criteria, but are not in the current flying build, which runs gate, registration, detector and scene classifier only. They consume the GNSS and IMU stream, and the trials conducted so far have been indoor bench and short-range tests where no GNSS fix is available—correctly, since the receiver reports no fix rather than a fabricated one. Integrating them is a wiring task against interfaces that already exist, not new research, but it is unmeasured in flight and is reported as such.

Finally, the system tracks the confidence of a *ground position* rather than the identity of a person. It has no re-identification capability comparable to the colour-histogram Bayesian matcher of [1], so two sightings of the same individual at different locations are two targets.

C. Future work

Three items follow directly. Wiring the registration-derived IoU and centroid-distance features into (10) is now unblocked and should recover a meaningful part of the fusion head's missing capacity. Adopting the sliding-window geometric OR-merge of [1] is achievable with the calibration in hand. And gate recall must be characterised on real thermal imagery over rubble before any coverage claim is made.

IX. CONCLUSION

We have described and measured a search-and-rescue perception payload that runs entirely on a 1 GB single-board computer with a 768-pixel thermal sensor. The organising principle is that the expensive model should never see a whole frame: a robust z -score test costing approximately nothing nominates two to six regions, and only those are examined. The consequence is a worst-case 177.5 ms of a 250 ms budget in 95.2 MB of memory, on hardware for which the prior literature reported 5.5 s per frame for the equivalent task.

Along the way we report three results with implications beyond this system: that the conventional stride-4 small-object head loses

to the standard head when the regime is information-limited rather than sampling-limited; that full INT8 quantisation can degrade a DFL-based detector to 31.9% of reference accuracy while presenting no observable symptom, and that quantising weights alone recovers 99.2% at identical file size; and that sub-pixel automatic correspondence detection makes cross-modal calibration tractable at a 45:1 resolution ratio where manual annotation is not, reaching 0.506 thermal pixels of reprojection residual. The complete system has been flown end-to-end with zero errors and no thermal throttling.

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All quantitative results in this paper were measured on the hardware described in Table I using the evaluation scripts in the project repository at commit [a6b8d47](#); none are reproduced from third-party reports except those in Table IX, which are attributed to their sources. Figures 2, 5 and 9 are unretouched captures from the payload.